SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***learning algorithms using backprop gradient descent optimization techni Maximum LikelihoodEstimation for real-world problems.***	***eptrons, multilayer***	***mental concepts of machine learning and neural networks to design and analyze perceptrons, forward propagation,***
Student Name:	Anand paul .D	**Reg. No.:**	192425207
Section / Batch:	2024	**Date:**	06-08-2026

Code of Conduct

I D. Anand Paul (192425207) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: bhupalreddy

QUESTIONS: 5 Total Marks: 50

Annexure A

Scenario-Based Assignment Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex real world datasets.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question:

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

6. Restricted Boltzmann Machines (RBM)

An online music streaming platform wants to improve its recommendation system by learning hidden relationships between users and songs from historical listening data without using labeled information.

Question:

Explain how a Restricted Boltzmann Machine (RBM) can be used in this recommendation system. Describe the architecture of an RBM, its working principle, training process, and discuss its advantages and limitations in real-world recommendation applications.

7. Autoencoders

A manufacturing company installs hundreds of sensors on industrial machines to continuously monitor their health. Most of the collected sensor data represents normal machine behavior, while faulty conditions occur very rarely. The company wants an AI system that can automatically identify abnormal machine behavior without requiring labeled fault data.

Question:

Discuss how an Autoencoder can be used for anomaly detection in this scenario. Explain the architecture of an Autoencoder, the roles of the encoder, bottleneck layer, and decoder, and describe how reconstruction error helps detect abnormal conditions.

8. Deep Learning Applications

A smart city project aims to integrate multiple AI-based services, including face recognition for security, speech recognition for public information kiosks, medical image analysis in hospitals, autonomous traffic monitoring, and intelligent language translation for tourists.

Question:

Explain how Deep Learning can be applied in each of these services. Discuss the advantages of Deep Learning in solving these real-world problems and identify the challenges that may arise during deployment.

9. Selecting the Appropriate Deep Learning Technique

A technology company is planning to build an AI-powered platform with the following features:

- Image classification for uploaded photos
- Customer recommendation system
- Data compression for cloud storage
- Feature extraction from unlabeled data

The management asks the AI team to recommend suitable Deep Learning techniques for each task.

Question:

Identify the most appropriate Deep Learning models or techniques for each feature. Justify your recommendations by explaining the working principles and advantages of Representation Learning, Autoencoders, Restricted Boltzmann Machines, and other relevant Deep Learning concepts.

10. Designing a Deep Learning Solution

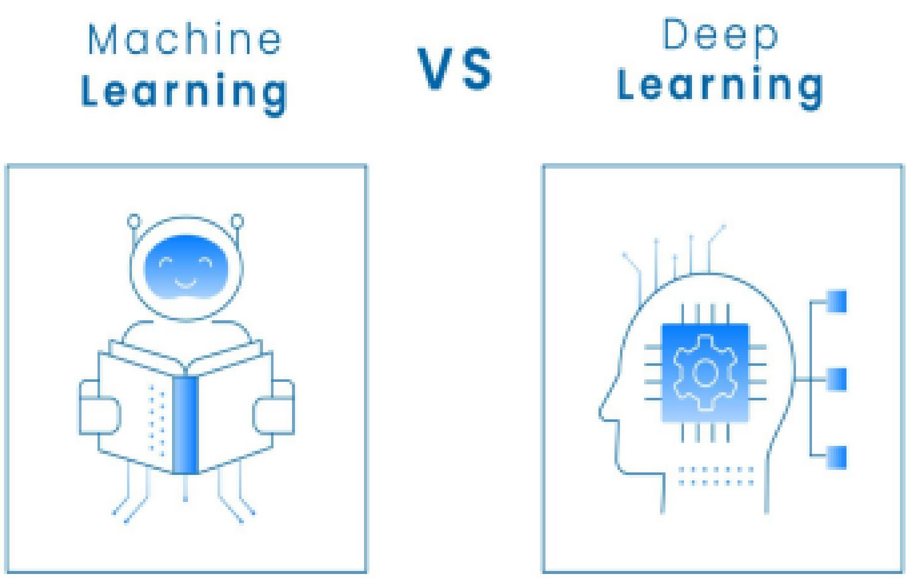
A national disaster management agency wants to build an intelligent emergency response system capable of analyzing satellite images, identifying damaged buildings, classifying disaster severity, and providing rescue recommendations. The system must process large amounts of image data accurately and efficiently.

Question:

Design a Deep Learning-based solution for this application. Explain whether Machine Learning or Deep Learning is more suitable, discuss the importance of Representation Learning, recommend appropriate activation functions for the network, and suggest whether unsupervised techniques such as Autoencoders or Restricted Boltzmann Machines could improve the overall performance of the system. Support your answer with appropriate reasoning and real-world examples.

1. Machine Learning vs. Deep Learning for Pneumonia Detection

When shifting from traditional Machine Learning (ML) to Deep Learning (DL) for complex image classification tasks like analyzing chest X-rays, the development paradigm fundamentally changes, particularly in how data is processed and features are understood.



Manual vs. Automated Feature Extraction. Source: Abdul Basit Noohani / Getty Images

The Deep Learning Paradigm Shift While traditional ML models hit a performance plateau—meaning their accuracy stops improving even if you feed them millions of images— Deep Learning architectures scale logarithmically with data. By utilizing Convolutional Neural Networks (CNNs), the team no longer needs to tell the model *what* to look for. Instead, they feed the raw pixel data directly into the network, allowing the model to automatically discover and learn the most critical visual markers of pneumonia on its own.

Comparison in Context

Feature	Traditional Machine Learning	Deep Learning (e.g., CNNs)
Feature Extraction	Requires manual, domain expert engineering (e.g., edge detection filters, texture analysis).	Automated hierarchical extraction. Learns low-level features (edges) to high-level concepts (opacity indicating pneumonia).
Data Requirements	Performs acceptably on smaller datasets. Plateaus in accuracy even if given millions of images.	Thrives on massive datasets. Accuracy continues to scale logarithmically as the dataset grows to millions of images.
Computational Resources	Low to moderate. Can often be trained on standard CPUs.	Extremely high. Requires specialized hardware (GPUs/TPUs) for training due to millions of parameters.
Accuracy on Images	Limited by the quality of the handcrafted features. Struggles with spatial translations and complex variations.	State-of-the-art accuracy. Convolutional Neural Networks (CNNs) inherently capture spatial hierarchies and complex non-linear patterns.

Justification for Deep Learning

Deep Learning is the superior choice for this healthcare application because the team now has **millions of medical images**. Traditional ML cannot effectively harness a dataset of this magnitude, nor can manual feature engineering capture the incredibly subtle, complex visual markers of pneumonia as accurately as a deep neural network. While real-world implementation will require heavy computational infrastructure (cloud GPUs) and rigorous validation to ensure the model isn't learning spurious correlations (like hospital tags on X-rays instead of lung tissue), the performance ceiling of a CNN makes it indispensable for life critical medical diagnostics.

2. Representation Learning in Retail Data

Representation Learning is a set of techniques that allows a system to automatically discover the representations (or features) needed for feature detection or classification from raw data, without human intervention. Instead of a data engineer manually writing rules to group data, the neural network learns to map the raw data into a dense, lower-dimensional space (an embedding) where similar patterns are grouped together.

How it Solves the Retail Problem

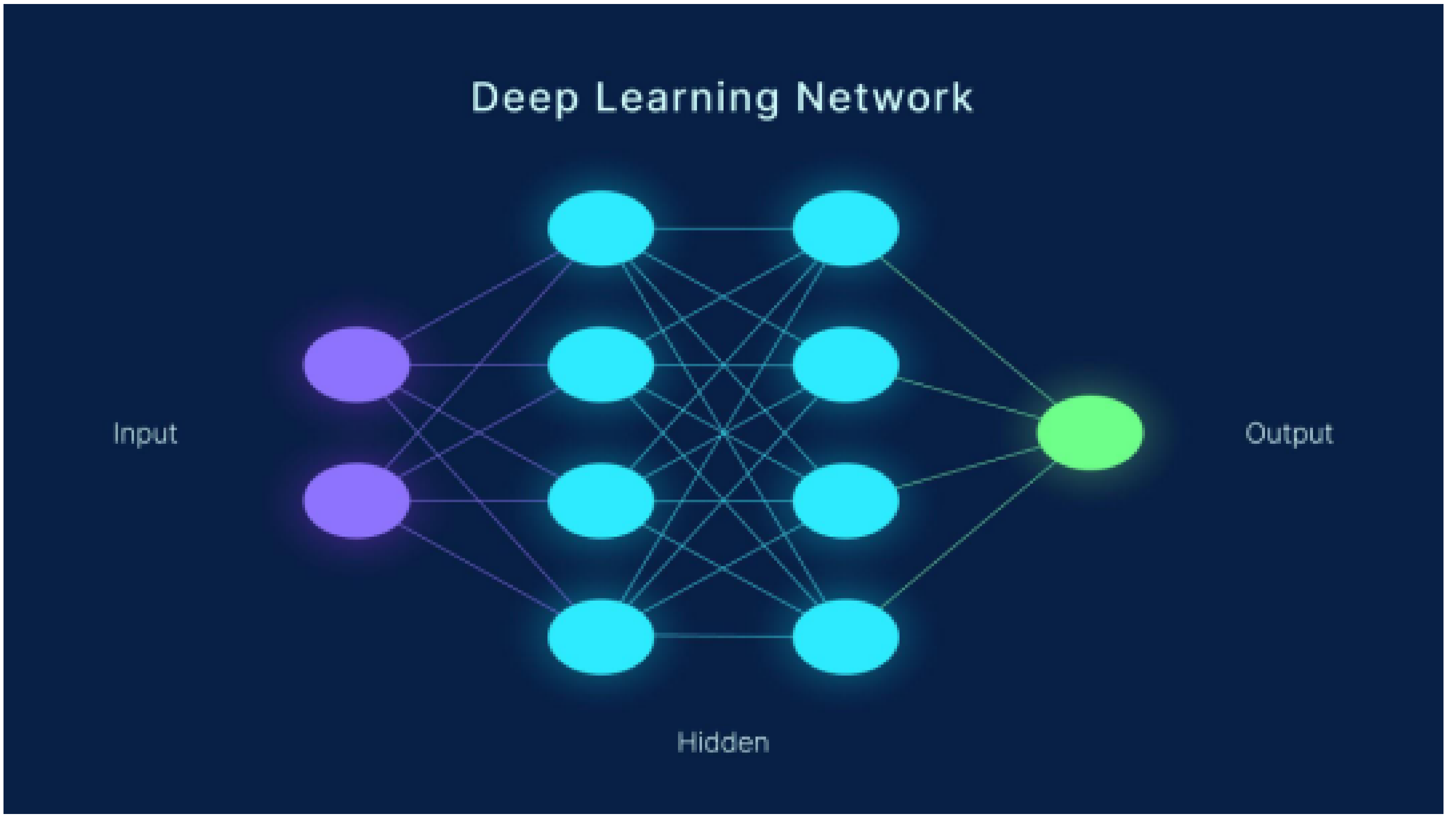
In this scenario, the retail company has millions of raw transactional records (e.g., timestamps, item IDs, quantities). Traditional feature engineering would require analysts to guess which features matter (e.g., "purchases in the last 7 days", "average spend per category"). Representation learning bypasses this bottleneck. Using architectures like Autoencoders or Recurrent Neural Networks (RNNs) on sequential purchase histories, the model will naturally learn latent variables—such as a user's hidden preference for organic food or seasonal buying habits—simply by attempting to predict their next purchase.

Advantages over Traditional Feature Engineering

- **No Domain-Expert Bottleneck:** It removes the need for humans to hypothesize and hard code features, which is slow and prone to human bias.
- **Captures Hidden Interactions:** Traditional features are usually linear or simplistic. Representation learning captures highly complex, non-linear interactions across thousands of sparse variables (like specific product pairings) that humans would never manually identify.
- **Scalability:** As the company adds new product lines or transaction types, the model dynamically learns the new representations upon retraining, rather than requiring an engineer to write new feature-extraction code.

3. Width vs. Depth of Neural Networks for Digit Recognition

When designing a neural network for image data (like handwritten digits), the architectural debate often centers on **Width** (the number of neurons within a single layer) versus **Depth** (the number of stacked hidden layers).



Architecture of a Deep Neural Network. Source: nuddss / Getty Images

Architectural Analysis

- **Wide and Shallow Network (One layer, thousands of neurons):**
  - *Feature Learning:* It attempts to learn a direct mapping from raw pixels to the digit in a single step. It essentially acts as a massive template matcher, memorizing specific pixel configurations.
  - *Computational Complexity:* Creates an enormous parameter matrix in a single layer, which is computationally expensive to multiply and highly prone to overfitting because it memorizes the training data rather than generalizing concepts.
- **Deep and Narrow Network (Multiple layers, fewer neurons):**
  - *Feature Learning:* Learns hierarchically. The first layer might learn simple edges and curves. The second layer combines those edges into loops or lines. The deeper layers combine loops and lines to recognize actual digits (like an '8' or a '4').
  - *Generalization & Performance:* By forcing information through multiple, smaller layers, the network creates a more robust, generalized understanding of what makes a digit, making it resilient to changes in handwriting style, size, or position.

Recommendation

The **deeper network** is strongly recommended. Theoretical computer science (specifically circuit complexity) proves that deep networks can represent complex functions with exponentially fewer parameters than shallow networks. For computer vision tasks like handwritten digit recognition, hierarchical feature extraction is critical, making deep architectures drastically superior in both generalization and parameter efficiency.

Shift from Dense to Convolutional Layers (CNNs)

If the team builds a standard deep Feedforward Network (Multilayer Perceptron), they have to flatten the 2D image into a 1D line of pixels. This destroys the spatial relationships between neighboring pixels.

- **The Recommendation:** Instead of standard dense layers, use **Convolutional layers**. CNNs slide small filters (like 3x3 pixel grids) over the image, allowing the network to understand 2D spatial structures (curves, loops, edges) regardless of exactly where the digit is drawn on the canvas.

Use a "Pyramid" Architecture

Deep networks shouldn't be the same width from start to finish. The most efficient design for computer vision is a pyramid shape.

- **Early Layers (Wide spatially, shallow depth):** Capture large spatial dimensions (the raw image) but only extract a few feature maps (like basic edges).
- **Deeper Layers (Narrow spatially, massive depth):** Use pooling layers to shrink the spatial dimensions (compressing the image size), while drastically increasing the number of feature channels (extracting hundreds of complex patterns like "closed loop at the top" for a 9 or 8).

4. Evaluating Activation Functions

The issues observed by the team—slow convergence, vanishing gradients, dead neurons, and unstable outputs—are direct consequences of misapplied activation functions.

Layer-by-Layer Evaluation

➢ **Sigmoid and Tanh (Avoid in Deep Hidden Layers)**

• *Characteristics:* Both squash inputs into a small range (Sigmoid: 0 to 1; Tanh: -1 to 1). • *The Problem:* When dealing with deep networks, their derivatives become extremely small (close to zero) for large or small inputs. This causes the **Vanishing Gradient** problem, which is exactly why the team is experiencing "slow convergence" and models failing to learn in early layers.

• *Verdict:* Do not use these in the hidden layers of a deep CNN.

➢ **Standard ReLU (Use cautiously in Hidden Layers)**

• *Characteristics:* f(x)=max(0,x). It passes positive values directly and outputs zero for negative values.

• *The Problem:* It solves the vanishing gradient for positive values, allowing fast convergence. However, it completely blocks negative values (gradient is zero). If a large gradient update pushes a neuron's weights so that it always outputs a negative number, the neuron permanently dies—causing the **"Dead Neurons"** issue observed by the team.

➢ **Leaky ReLU and ELU (Best for Hidden Layers)**

• *Characteristics:* Leaky ReLU allows a small, non-zero gradient (e.g., 0.01x) when the input is negative. ELU (Exponential Linear Unit) smoothly curves into negative values. • *The Solution:* Both mathematically eliminate the "dead neuron" problem by ensuring gradients can still flow backwards even when the neuron receives a negative input, keeping the network healthy and convergence fast.

• *Verdict:* **Use Leaky ReLU or ELU in all hidden layers.**

➢ **Softmax (Mandatory for Final Layer)**

• *Characteristics:* Converts raw output scores (logits) into a normalized probability distribution (all outputs sum to 1.0).

5. Unsupervised Training of Neural Networks

For a social media platform possessing billions of unlabeled images, manual annotation is economically unfeasible. Unsupervised (or Self-Supervised) training allows the neural network to learn the underlying structure and rich representations of the data without needing explicit "cat" or "dog" labels.

The Learning Process

Instead of learning a mapping from an image to a human-provided label, the network uses the data *itself* as the supervisory signal.

1. **Autoencoders:** The network is forced to compress an image into a tiny, dense bottleneck

layer (encoding) and then reconstruct the original image perfectly from that bottleneck (decoding). To succeed, it *must* learn the most critical structural features (colors, shapes, textures) of the images.

2. **Contrastive Learning (e.g., SimCLR):** The network takes an image, creates two augmented versions of it (e.g., one cropped, one color-shifted), and is trained to push their mathematical representations closer together, while pushing the representations of *different* images further apart.

Advantages over Supervised Learning

- **Unhindered by Data Bottlenecks:** It leverages the entirety of the billion-image dataset, creating a foundational model with an incredibly deep understanding of visual geometry and lighting.
- **Transferability:** Once the unsupervised model learns these dense representations, it can be fine-tuned on a drastically smaller *labeled* dataset for specific tasks (like detecting inappropriate content or tagging friends) with state-of-the-art accuracy. It solves the cold start problem of deep learning by extracting value from raw data scale.

			wi vis
Communication	15	Highly engaging, confident, and audience-focused delivery	C or ar pr cor n
Professionalism	10	Exemplary professionalism, leadership, and ethicalawareness	Pr c w e te c n

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exempl
Technical Content Accuracy	30	Highly accura indepth, a insightful technical explanation
Problem Identification	25	Problem rigorously analyzed us appropriate a advanced methods
Use of Tools	20	Advanced tools well analyzed a and highly effective visua

Faculty In-charge Course Coordinator Program Director
Signature: Signature: Signature:
Date: Date: Date: